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ARTICLE

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Improving the specific wear rate and coefficient of friction of polyamide 6 polymer and its composite by adding wax under self-operation conditions

Verbesserung der spezifischen Verschleißrate und des Reibungskoeffizienten des Polymers Polyamid 6 und seines Verbundstoffes durch Zugabe von Wachs unter Selbstbetriebsbedingungen

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Abstract
Polyamide polymers are widely used in tribological areas in a wide range of industries. The purpose of this study is to extend the tribological life of this material. For this purpose, composite materials were produced by adding solid lubricants such as graphite and wax to the polyamide 6 (PA6) matrix at different rates, and the synergistic effect of these composites on tribological behavior was investigated. Tribological experiments were carried out under dry sliding conditions and on themselves. The polyamide 6/graphite/wax composites were first produced in the form of granules in a twin-screw extruder, and then test samples were molded using the injection molding technique. Wear tests were performed using a pin-on-disc wear device. Tribological tests were carried out at a sliding speed of 0.5 $m\,s^{-1}$ and under loads ranging from 10 N to 250 N. Following the experiments, the friction coefficient, specific wear rate, and surface wear were determined under the condition of the materials working on each other. The results showed that the lowest specific wear rate and friction coefficient were obtained for the polyamide 6 composites containing 6 % wax and 15 % glass fiber working on each other.

KEYWORDS
composite, friction, graphite, polyamide 6, wax, wear

Abstract
Polyamid Polymere werden in tribologischen Bereichen in einer Vielzahl von Branchen eingesetzt. Ziel dieser Studie ist es, die tribologische Lebensdauer dieses Werkstoffes zu verlängern. Dazu wurden Verbundwerkstoffe hergestellt, indem der Polyamid 6 (PA6)-Matrix Festschmierstoffe wie Graphit und

Wachs in unterschiedlichen Anteilen zugesetzt und die synergistische Wirkung dieser Verbundwerkstoffe auf das tribologische Verhalten untersucht wurde. Es wurden tribologische Versuche unter Trockengleitbedingungen und mit sich selbst durchgeführt. Die Polyamid 6/Graphit/Wachs-Verbundwerkstoffe wurden zunächst in Form eines Granulats in einem Zweischneckenextruder hergestellt und dann die Prüfkörper im Spritzgussverfahren geformt. Verschleißtests wurden unter Verwendung einer Stift-Scheibe-Verschleißvorrichtung durchgeführt. Tribologische Tests wurden bei einer Gleitgeschwindigkeit von 0,5 $m\,s^{-1}$ und unter Belastungen von 10 N bis 250 N durchgeführt. Im Anschluss an die Versuche wurden Reibwert, spezifischer Verschleiß und Oberflächenverschleiß unter den Selbstbetriebsbedingungen der Werkstoffe bestimmt. Die niedrigste spezifische Verschleißrate und der niedrigste Reibungskoeffizient unter Selbstbetriebsbedingungen wurden mit den Polyamid 6-Verbundwerkstoffen erzielt, in denen 6 % Wachs und 15 % Glasfaser enthalten waren.

SCHLÜSSELWÖRTER
Graphit, Polyamid 6, Reibung, Verschleiß, Verbundwerkstoff, Wachs

1 | INTRODUCTION

The production of materials that have lower friction and wear properties during contact between mechanical parts is one of the main goals of industrial studies. The development of new materials with optimum properties in this field continues to attract interest in industrial and scientific research [1]. Pure polymeric materials have lower mechanical, thermal, and tribological properties compared to metal and ceramic materials, limiting their use [2, 3]. Mechanical, tribological, and thermal properties of pure polymers need to be improved for wear and friction-related applications in the machinery industry [4]. The improved and more economic performance of polymer-based composite materials are rapidly replacing traditional materials such as steel and ceramics. Polymer-based composite materials are used in place of traditional materials in many different fields such as aerospace, aviation, sports equipment, construction, automotive, household appliances, and the machinery industry [5]. In the machinery sector, many plastic materials work either together with metal materials or through contact with polymer materials. Previous research has determined that the wear as a result of polymer-polymer contact is much more severe than the wear as a result of polymer-metal contact. At the same time, determining the wear resistance and friction coefficient of materials is one of the most basic parameters in tribology. The coefficient of friction is the resistance one material has to moving relative to another. A low coefficient of friction is often one of the best indicators in tribology [6]. Many kinds of additives in different forms and properties are used in the industry to improve the tribological, mechanical, and thermal properties of polymer-based materials. Reinforcements such as glass fiber, carbon fiber and aramid fiber and filling materials such as talc, calcite, kaolin, glass ball, wollastonite are generally used to increase mechanical properties. Solid lubricants such as carbon, Teflon, graphite, molybdenum disulfide and wax are added to the matrix especially to improve their tribological properties [7]. At low sliding speeds, the service life of polyamide 6 is 2 times to 10 times longer than metal-based materials. In addition, by adding solid lubricants to polyamide 6, polymer-based composites with superior tribological properties are produced [8]. Polyamide 6 is widely used in drinking water transportation and storage applications due to the good chemical resistance of polyamide engineering polymers and in the construction of machine parts thanks to its superior mechanical properties. In addition, the fact that they are 7 times lighter than steel provides a very important advantage to polyamide materials [9]. However, the dehumidification feature of polyamide 6 increases its resistance against vibrations and impacts during use, while complicating dimensional stability during the production of parts. It also has disadvantages such as notch sensitivity, low impact strength at low temperatures, and low ultraviolet resistance [10]. Graphite is very soft, oily to the touch and can be bent into thin sheets. While it burns at 600 °C–670 °C in an oxygenated environment, it can go up to 3500 °C without burning in normal air environments. Graphite melts at 3927 °C, but at this temperature, it turns directly into the gas, not liquid. While the bonds within the planes formed by the carbon atoms of graphite are very strong, the bonds between the planes are quite weak. The reason graphite is a good lubricant is that these planar carbon sheets slide easily over each other [11, 12].

Additives such as wax are used as lubricants and homogenizers in polymer materials. Thus, they improve the general thermomechanical properties of polymer-based composites [13]. The number of carbon atoms in the chains of paraffin wax ranges from 18 to 50 (C18 to C50). The increased length of the carbon atom chains increases the molecular weight and results in a higher melting temperature [14]. By blending wax with polymers, materials with a controlled structure and phase change can be produced. A polymeric matrix compactly secures the paraffin wax even after it has melted and prevents wax leakage. Such materials are easily shaped and impart unique properties to the polymeric phase [15]. While polymer/wax mixtures are produced in the industry primarily for the production of polymer materials in the field of tribology, they are also made to be used for a number of products like hot melt adhesives, laminates, coatings, casting waxes, and rubber additives [16]. The wear properties of polymers are improved through the addition of solid lubricants to produce polymer composites [17–30].

The addition of graphite to polymers, including polyamide 6, leads to a decrease in the coefficient of friction, specific wear rate, and tensile and impact strength, while leading to an increase in flame resistance and wear resistance [27, 31–33]. As the ratio of graphite in the graphite/polymer composites increases, the coefficient of friction, specific wear rate, tensile and impact strength, and elongation at break decrease, while increasing hardness values and the modulus of elasticity increase [27, 34–36]. The reduction in the coefficient of friction and wear rate with the addition and increase of graphite in graphite/polymer composites can be explained by the fact that the graphite has a layered structure and the layers slide against each other during sliding, forming a thin film layer of graphite on the opposite disc material surface. The coefficient of friction and wear rate of graphite-added polymer composites have been found to decrease with increasing applied load and increase with increasing sliding speed, however these findings have not been consistent [34–37]. Overall, the addition of graphite to polymers leads to an overall enhancement of tribological properties. The addition of a 10 wt.% graphite filler decreases the wear rate of carbon fabric reinforced epoxy composites due to the graphite filler forming a thin and uniform film layer on the opposite disc material [38]. The addition of a graphite filler also increases the wear resistance and decreases the coefficient of friction of epoxy matrix composites due to the filler increasing film layer thickness and uniformity on the surface of the opposite material [39]. The solid lubricating property of graphite and the effect of reducing the coefficient of friction are explained by the strong covalent bond between carbon atoms in the basal planes and the weak (Van der Waals) forces between adjacent planes and its layered structure. This weak bond can be easily broken during sliding, and the released material forms a thin tribolayer at the contact surface, which reduces the contact surface.

The crystallinity ratio of the polymer also effects the wear and friction behavior of the polymer. The degree of crystallinity of graphite and wax filled composites decreases with the addition of graphite and wax filler to the polymer matrix [40]. The degree of crystallinity of the glass fiber reinforced polyamide 6 composites decreases with the increase in the expanded graphite ratio [41].

The properties of polymer materials with a semi-crystalline structure are closely related to the thermo-mechanical history of the material and the process conditions. The degree of crystallinity of the material, Xc, can be explained by the texturing and chain orientation in the structure. The intrinsic properties of crystalline and amorphous structures are often very different. These structures affect the macroscopic behaviour of the material [42–44]. Rapid cooling of the molten polymer in the mould causes the degree of crystallization to be lower than the maximum crystallization value in polymer materials with low crystallization kinetics. Low crystallisation kinetics implies that there should be sufficient temperature and time for nucleation and growth of nuclei in the polymer melt. For polymers with a glass transition temperature above room temperature, crystallisation can theoretically be completely prevented by extremely rapid cooling and/or the solidified polymer will be fully or partially amorphous. In polymers with glass transition temperature above room temperature, crystallization can be completely prevented by rapid cooling. One of the polymers with glass transition temperature above room temperature is polyamide 6 polymer. As a result of the thermal processes to which polyamide 6 polymer material is exposed, the two most common phases are alpha and gamma phases. The monoclinic alpha form is the most stable phase and is usually obtained from the quasi-essential melt at crystallization temperatures $Tc > 150\,°C$ [45, 46–48]. Pseudo-hexagonal gamma-crystals are obtained from quiescent melts at $Tc < 150\,°C$ [45, 46–49].

The thermal history of the material actually affects both the mechanical and tribological properties of the material. The wear resistance of glass fibre reinforced polyamide composites is significantly affected by the thermal history of the composite [50]. In other words, slow cooling of the part/material in the mould leads to increased crystallinity and high crystallinity ratio. Slow cooling will also provide higher thermal stability, higher density and harder matrix material than fast cooling. As a result of increased crystallinity or slow cooling, the interfacial properties (thermal resistance and density) and hardness of the material are also improved. In graphite/polyamide 6 composites, it results in an increase in wear resistance by about 28 % and a reduction in friction by about 14 %. Slow cooling gives polyamide 6 high crystallinity or high crystallinity ratio, resulting in higher thermal stability, higher density and stiffer matrix in composites compared to fast cooling. It also improves interfacial properties and composite stiffness. Since thermal decomposition, abrasion and adhesion dominate the wear mechanisms, these improvements led to higher wear resistance and higher Pressure Velocity (PV) service limit in slow cooled graphite/polyamide 6 specimens than in fast cooled specimens.

In many industrial areas, polymer/polymer combinations attract more attention due to their low cost and lightness properties, and many machine parts (such as gear mechanisms) used in the industry make use of polymer/polymer pairs. For polymer/polymer pairs, both the coefficient of friction and specific wear rate decreases with increasing applied load, and they both increase with increasing sliding speed. In contrast, for polymer/steel pairs, the wear rate increases with increasing applied load and the coefficient of friction decreases with increasing sliding speed [51, 52]. Polymer/polymer pairs also have been found to have a higher specific wear rate and coefficient of friction relative to polymer/steel pairs [51].

The correct selection of the polymer materials used in the field of tribology, which are used in many areas in the industry, will lead to longer service life. For this purpose, the wear properties of polyamide 6 polymer and polyamide 6 composites with 5 %, 10 % and 15 % graphite additives were investigated. When it was determined that the friction coefficient and wear rate of these composites were high, 2 % and 6 % wax solid lubricant was added to the polyamide 6–15 % graphite composites and the effect of the wax additive on the polymer/polymer pairs was investigated. The test results showed a significant decrease in both the friction coefficient and the specific wear rates due to the wax additive. Also, the load-carrying capacity of the wax-added composites was 7.33 times greater compared to the load-carrying capacity of the pure polyamide 6 and the 15 % graphite added polyamide 6 composites.

2 | EXPERIMENTAL STUDY

The polyamide 6 matrix material used in this study has the commercial code Domomid 6 NC01 and was obtained from Domo Chemicals in Belgium. The wax used as a solid lubricant is a solid lubricant, that has the commercial code Licowax E flake and is approximately 1.02 $g\,cm^{-3}$ in density, which can provide internal and external lubrication to high heat stability polyamide engineering polymers. Licowax E flakes is an ester of montanic acids with multifunctional alcohols and was obtained from Clariant in a flake form. Graphite was obtained from Focus Co. Company (Istanbul, Turkey). Wax and graphite added polyamide 6 polymer composite granules were prepared using a twin-screw extruder (NR II-75 type) at a heater temperature range of 220 °C to 245 °C. Mixtures in the form of granules were heated at 80 °C for 4 hours before injection molding. The test samples were produced using the injection molding technique, within the temperature range of 220 °C to 250 °C. Graphite and wax additive filled polyamide 6 compound formulations have been developed to reduce wear rate and friction coefficient of many machine elements working on each other, such as gear mechanisms. In the study, six different samples were created with the following formulations; pure polyamide 6, polyamide 6–5 % graphite, polyamide 6–10 % graphite, polyamide 6–15 % graphite, polyamide 6–15 % graphite-2 % wax, and polyamide 6–15 % graphite-6 % wax. Samples, 6 mm in diameter and 50 mm in length, were produced by injection molding for the mechanical and wear tests. Mechanical tests were performed on specimens prepared in accordance with the Standard Test Method for Tensile Properties of Plastics (ASTM D638). Zwich Z020 brand test machine was used for tensile test. At least five test specimens were used in the tensile test and average values were given. Tribological tests were carried out in a pin-on-disc wear device in accordance with the Standard Test Method for Wear Testing with a Pin-on-Disk Apparatus (ASTM G99). The tests were carried out at 23 °C room temperature and dry sliding conditions. It was carried out at a sliding speed of 0.5 $m\,s^{-1}$ and a sliding distance of 1000 m under loads ranging from 10 N to 250 N. The tribology studies of the test specimens used in the experiments were continued until the maximum load at which the composite materials could work without plastic deformation. Test temperature, sliding distance, sliding speed and load were used as the tribology test parameters, Table 1.

Before each test, the surfaces of the polymer disc on which the wear test was performed and the sample

TABLE 1 Test materials and conditions.

Material	Test temperature (°C)	Sliding distance (m)	Sliding speed ($m\,s^{-1}$)	Load (N)
Pure polyamide 6	23 ± 2	1000	0.5	10, 20, 30, 50,75, 100, 200, 250
Polyamide 6–5 % graphite				
Polyamide 6–10 % graphite				
Polyamide 6–15 % graphite				
Polyamide 6–15 % graphite-2 % wax				
Polyamide 6–15 % graphite-6 % wax				

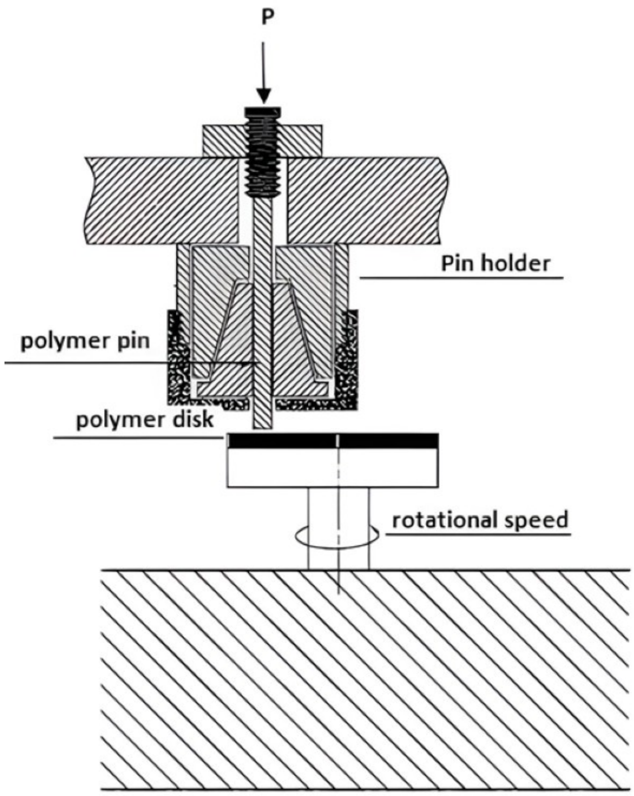

FIGURE 1 Schematic view of the pin-on-disc device in which the wear tests were carried out.

surfaces of the pin used in the wear test were cleaned with acetone. The pin-on-disc device, in which the wear tests were carried out, consists of an electric motor, computer, load cell, pin holder, speed adjuster, and load holder parts, Figure 1. Samples produced for wear tests are attached to the pin holder. The speed adjuster ensures that the electric motor is rotated at the desired speed. The load to be used in the test is added to the pin arm and the wear test is initiated by leaving it on the disc. During the experiments, the friction force is measured with a load cell mounted on the loading arm. During the experiment, approximately 1500 data points were collected every 60 seconds and recorded in an Excel program on the computer. The friction coefficient was calculated by dividing the lateral force by the applied load using the formula in Equation 1.

$$\mu = \frac{F_S}{F_N} \tag{1}$$

In the formula above; μ: friction coefficient, F_S: lateral friction force, and F_N: wear load applied on the sample.

In order to find the weight loss due to wearing, the weights of the pin samples before and after the test were measured with a Precia brand precision balance, with 220 g capacity and 0.001 g sensitivity, and the weight loss (Δ_m) was calculated by subtracting the post-test weight (M_2) from the pre-test weight (M_1). It was calculated with the formula in Equation 2.

$$\Delta_m = M_1 - M_2 \tag{2}$$

The specific wear rate was calculated using the formula given in Equation 3.

$$K_0 = \frac{\Delta_m}{L\,\rho\,F} \tag{3}$$

In this formula; K_0: specific wear rate ($m^2\,N^{-1}$), L: slip distance in meters, ρ: density of the test samples (g cm^{-3}), and F: wear load in Newton (N).

Each experiment was repeated at least three times and the arithmetic average was taken.

To determine the thermal properties of composites H-DSC 60 Shimatzu brand H-DSC-60 model the differential scanning calorimetry (DSC) analysis was performed. The device was operated in the temperature range of 25 °C to 300 °C. The device heating rate was selected as 10 °C/min. Melting temperatures and enthalpy changes of composite materials were determined. For each composite material, two studies were carried out and the data from the second study were taken into consideration.

3 | EXPERIMENTAL RESULTS AND DISCUSSION

The degree of crystallisation of the polymer significantly affects the properties of plastic materials. If the degree of crystallisation of the polymer is high, the material becomes harder, stronger and more brittle. The degree of crystallisation (Xc) of polymer materials is calculated by using the formula in Equation 4 [53, 54].

$$Xc\ (\%) = \frac{\Delta H_{meas}}{(1-w)\ \Delta H_{lite}} * 100 \quad (4)$$

In the formula, w is the weight fraction of graphite filler, ΔH_{meas} is the measured enthalpy of melting of the polymer material and ΔH_{lite} is the enthalpy of melting of the 100 % crystalline material. Differential scanning calorimetry (DSC) was performed to determine the melting enthalpy of the materials used in the experiment. To calculate the degree of crystallinity, the melting enthalpy of polyamide 6 polymer, which is 100 % crystalline in the literature, was taken as 230 (±20) J/g [54–56]. Data of polyamide 6 and graphite/wax composites from differential scanning calorimetry (DSC) device are given, Figure 2. At the same time, the thermal analysis values of pure polyamide 6 polymer and polyamide 6–5 % graphite composite and polyamide 6/graphite/wax composites are given, Table 2. No significant change was observed in the melting temperature of the composites when compared with the melting temperature of the pure polyamide 6 polymer, Figure 2. The highest melting enthalpy was 96.72 J/g in pure polyamide 6 polymer. The melting enthalpy of polyamide 6–15 % graphite composite was determined as 39.56 J/g. Polyamide 6–15 % graphite-2 % wax, and polyamide 6–15 % graphite-6 % wax, melting enthalpy of composites were determined as 42.02 J/g, and 32.57 J/g, respectively. While the crystallinity of pure polyamide 6 was 42 %, the crystallinity of the composite decreased to 20.23 % with the addition of 15 % graphite to the polyamide 6 main matrix. By adding 2 % and 6 % wax to the 15 % graphite added polyamide 6 composite, the crystallinity of the composite increased slightly as 21.49 %. With the addition of 6 % wax, the crystallinity of the composite decreased to 16.65 %.

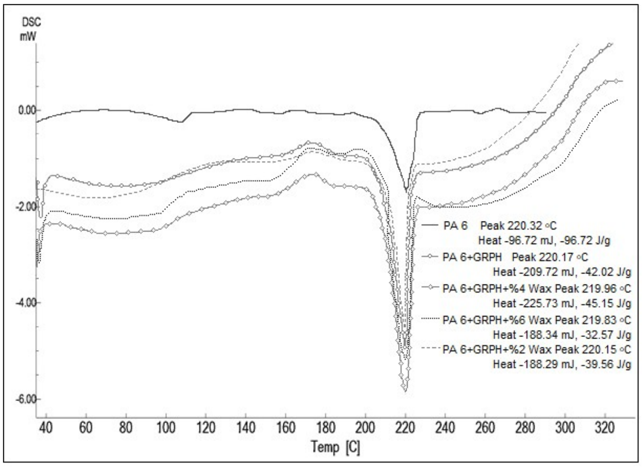

FIGURE 2 Differential scanning calorimetry (DSC) thermograms of polyamide 6 polymer, graphite and graphite/wax filled polyamide 6 composites.

TABLE 2 Differential scanning calorimetry (DSC) thermal analysis results of pure polyamide 6 polymer and graphite/wax filled polyamide 6 composites.

Materials	Tm (°C)	ΔHm (J/g)	Xc (%)
Pure polyamide 6	220.32	96.72	42
Polyamide 6–15 % graphite	220.15	39.56	20.23
Polyamide 6–15 % graphite-2 % wax	220.17	42.02	21.49
Polyamide 6–15 % graphite-6 % wax	219.83	32.57	16.65

The mechanical properties of graphite filled polyamide 6–15 % graphite composite, polyamide 6–15 % graphite-2 % wax and polyamide 6–15 % graphite-6 % wax composites used in the experiments, Table 3. The addition of graphite filler to the polyamide 6 main matrix caused an increase in the hardness of the composite, Table 3. When compared to pure polyamide 6, the rate of

TABLE 3 Mechanical properties of pure polyamide 6 polymer and graphite/wax filled polyamide 6 composites.

Materials	Hardness shore, D	Tensile strength, MPa	Tensile modulus, MPa	Elongation at break, %
Pure polyamide 6	75 ± 1	75 ± 1.13	3001 ± 101	7.09
Polyamide 6–15 % graphite	79 ± 0.5	67 ± 1.21	4607 ± 125	4.23
Polyamide 6–15 % graphite-2 % wax	78 ± 0.5	63 ± 1.02	4387 ± 93	4.0
Polyamide 6–15 % graphite-6 % wax	78 ± 0.5	55 ± 1.42	4216 ± 112	3.95

increase in hardness was approximately 5.33%. When wax was added to the polyamide 6–15% graphite composite, a very slight decrease in hardness values was detected. The rate of decrease in hardness was approximately 1.2%. The graphite filler added to the polyamide 6 polymer caused a decrease in the tensile strength of the composite material, Table 3. Compared to the tensile strength of pure polyamide 6 polymer, the tensile strengths of polyamide 6–15% graphite, polyamide 6–15% graphite-2% wax and polyamide 6–15% graphite-6% wax composite materials decreased. This decreasing ratio is 10.66%, 16% and 26.66%, respectively. Similarly, while the tensile strength values of the composites decreased, a decrease in the elongation at break values was also observed. However, a significant increase in tensile modulus values was found with the addition of graphite filler to the main matrix. With the addition of 15% graphite to the polyamide 6 main matrix, 53.5% increase in the tensile modulus of the composite was determined. With the addition of 2 wt.% wax and 6 wt.% wax to polyamide 6–15% graphite composite, 4.77% and 8.48% decrease in the tensile modulus of the composites were determined, respectively.

The friction coefficient of the pure polyamide 6 along the 1000 m sliding distance (under a load of 30 N and at a sliding speed of 0.5 $m\,s^{-1}$) is higher and the peaks in the friction graph are longer, Figure 3. The relationship between the friction coefficient and the sliding distance for all the samples (the pure polyamide 6 and the polyamide 6-graphite composites) at a sliding speed of 0.5 m/s and under a load of 30 N was determined, Figure 3a. The relationship between the friction coefficient and the sliding distance for each of the samples was determined separately, Figures 3b–d. The addition of 10% graphite to the polyamide 6 resulted in a decrease in the friction coefficient. With the increase of the graphite ratio in the polyamide 6 to 15%, the friction coefficient value decreased further and it was observed that the peaks in the friction graph were shortened. The coefficient of friction generally differs at the beginning and end of the experiment. If the abraded material can form a fast and continuous film on the disc surface, the friction coefficient will decrease in a short time and become more stable. Therefore, it can be said that the graphite added to the matrix reduces the friction coefficient value as a result of the solid lubricant effect and the friction becomes more

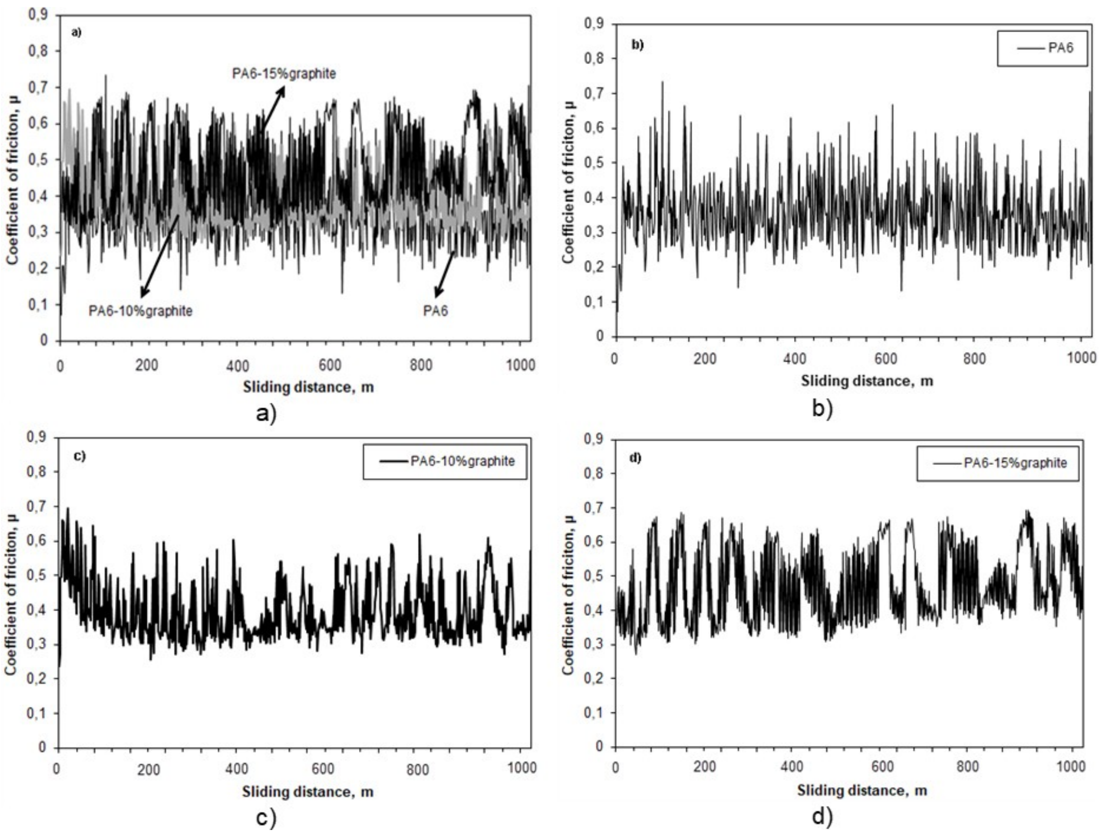

FIGURE 3 Graph showing the relationship between the friction coefficient and the sliding distance of a) Pure polyamide 6 (PA6) and polyamide 6-graphite composites, b) Pure polyamide 6, c) polyamide 6–10% graphite composites, d) polyamide 6–15% graphite composites (load: 30 N, sliding speed: 0.5 $m\,s^{-1}$).

stable along the sliding distance. The addition of graphite to polyamide 6 has a positive effect on reducing the coefficient of friction [23].

It was determined that the average friction coefficient of the composite containing polyamide 6 and 5% graphite is around 0.45 and this value constantly changes within the range of 0.35 to 0.85 along the sliding distance with to the addition of wax to the graphite-added polyamide 6 composite, the friction coefficient values were further reduced, Figure 4. While the average friction coefficient of the polyamide 6–15% graphite-2% wax composite was around 0.1, the peaks in the friction graph were very short. By increasing the wax ratio in the polyamide 6 to 6% by weight, the average friction coefficient value decreased to 0.09. The friction coefficient value was stable and almost unchanged along the sliding distance, Figure 4. This is consistent with previous studies [19]. It is thought that both the decrease in the friction coefficient value and the decrease in the peaks are due to the fact that the wax reduces friction resistance by forming a thin film layer on the wear surface.

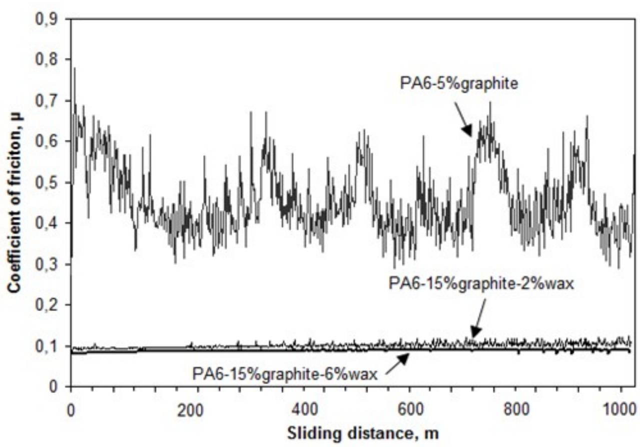

FIGURE 4 Graph showing the relationship between the friction coefficient and the sliding distance of pure polyamide 6 (PA6) and polyamide 6-graphite and polyamide 6-graphite-wax composites (load: 30 N, sliding distance: 0.5 m s^{-1}).

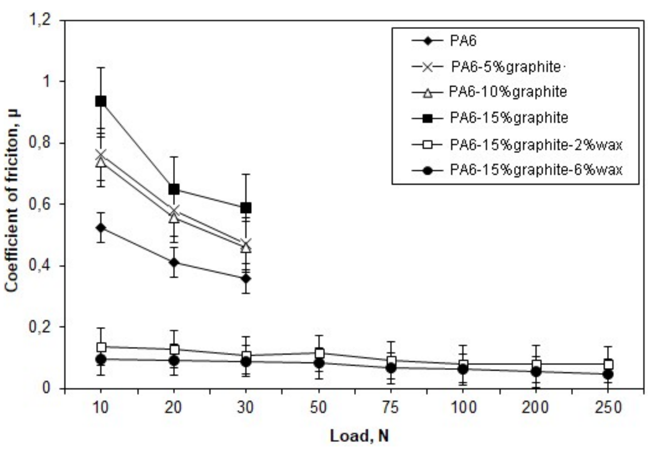

FIGURE 5 Graph showing the relationships between the friction coefficient values and the applied for the pure polyamide 6 (PA6) and the polyamide 6-graphite and polyamide 6-graphite-wax composites (sliding speed 0.5 m s^{-1}).

When the pure polyamide 6 and the polyamide 6-graphite composites were exposed to wear on themselves, the excessive mass loss occurred very quickly when the applied load exceeded 30 N (at a sliding speed of 0.5 m s^{-1}), Figure 5. Therefore, loads above 30 N were not used in the experiments for the pure polyamide 6 and the polyamide 6-graphite composites. With the increase of applied load from 10 N to 30 N, a significant decrease occurred in the friction coefficient of the pure polyamide 6 and the polyamide 6 + graphite composites. It was observed that the friction coefficient values of graphite added polyamide 6 composites at the rate of 5%, 10% and 15% graphite by weight were approximately 20%, 25%, and 36% lower, respectively, than polyamide 6 without additives. It was also observed that the friction coefficient values of the polyamide 6 composites with 15% graphite and 2% and 6% wax solid lubricants are lower by approximately 83% and 85%, respectively, compared to polyamide 6 without additives. In addition, when the load increased from 10 N to 250 N, the friction coefficients of the polyamide 6–15% graphite-2% wax and polyamide 6–15% graphite-6% wax composites do not decrease much. The friction coefficient values of the graphite and wax mixed polyamide 6 composites were very close to each other under all loads used in the tribology experiments. Among the materials used in the experiments, the highest friction coefficient value was obtained for the pure polyamide 6 polymer with a value of 0.98 under a 10 N load.

The lowest coefficient of friction was measured to be 0.07 for the polyamide 6–15% graphite-2% wax composite under a 250 N load. The friction coefficients of the graphite and wax added polyamide 6 composites were both lower and showed less change compared to the graphite added polyamide 6 composites with increasing load, Figure 5.

As is known in tribology, the actual contact area between the wear surfaces of materials increases with increasing applied load. However, since the increase in the actual contact area is much less than the increase in the applied load, the friction coefficient decreases with increasing load. These results are consistent with previous studies [21].

A significant decrease was observed in the specific wear rate with the addition of graphite to the polyamide 6 polymer relative to the pure polyamide 6 polymer, Figure 6. When the graphite additive ratio is high enough in the polyamide 6-based matrix, a thin film

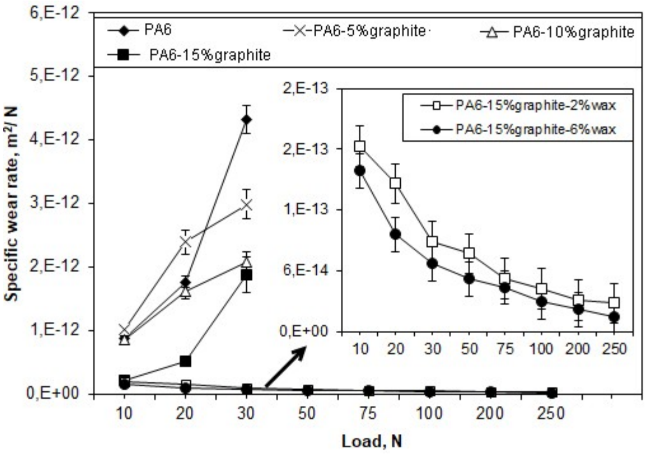

FIGURE 6 Graph showing the relationship between the specific wear rate and applied load for the pure polyamide 6 (PA6), and the polyamide 6-graphite, polyamide 6 graphite-wax composites (sliding speed 0.5 m s^{-1}).

layer containing graphite can form on the opposite surface. Since graphite has a layered structure and weak Van der Waals bonds between layers, the layers slide easily under sliding conditions, forming a film layer on the opposite disc surface. Graphite has the effect of reducing the contact between the opposing disc contact surface and the pin surface. This resulted in a decrease in both the friction coefficient and the wear rate of the graphite added polyamide 6 composites, Figures 5, 6. By increasing the applied load from 10 N to 30 N, an increase was observed in the wear rates for both the pure polyamide 6 and the polyamide 6/graphite composites. Experiments for the graphite-added polyamide 6 composites under loads over 30 N were not carried out due to excessive wear. For the pure polyamide 6, the increase in the specific wear rate was around 80% when the load was increased from 10 N to 30 N.

An increase of 50% was observed in the polyamide 6–5% graphite composite, and approximately 25% in the polyamide 6–10% graphite and the polyamide 6–15% graphite composites. In line with this data, it can be said that the graphite added to the polyamide 6 reduces the specific wear rate. The results obtained are in agreement with previous studies in the literature [27, 31–37]. The wear tests of the composites containing graphite and wax were able to work safely at a sliding speed of 0.5 m s^{-1} and under a load of 250 N, Figure 6. By adding wax and graphite to the polyamide 6 polymer matrix, the safe load carrying capacity increased from 30 N to 250 N. In other words, the safe load carrying capacity of polyamide 6-graphite-wax composite increased 7.33 times with the addition of graphite and wax. This provides the flexibility/advantage of working with higher loads on machine elements exposed to wear in the industry. A significant decrease in the specific wear rate of the wax-containing composites was observed in the 50 N to 250 N load range. By increasing the applied load by 400%, the specific wear rate decreased by 60% in the polyamide 6–15% graphite-2% wax composite, and by approximately 65% in the polyamide 6–15% graphite-6% wax composite. This is consistent with previous literature [28].

Wear marks are clearly visible on the pin and disc surfaces of the pure polyamide 6 polymer and the graphite filled polyamide 6 composites, Figure 7. Pure polyamide 6/pure polyamide 6 disc pairs have deep and wide wear marks with wavy wear marks on the contact surfaces, Figure 7a. Depending on the graphite ratio, deep and wide wear marks were observed on the contact surface of graphite filled polyamide 6/ graphite filled polyamide 6 polymer disc pairs, Figure 7b–d. For 5 wt.% and 10 wt.% graphite filled polyamide 6 composites while the wear marks were wider and deeper, fewer and narrower wear marks were observed in the 15 wt.% graphite filled polyamide 6 composite pair. It can be said that the wear is severe in graphite filled polymer composites and the abrasive wear mechanism is more effective.

It has been determined that the wear surfaces of the polyamide 6 + graphite composites are brighter. It is thought that this is caused by the graphite constantly breaking off during wear due to the weak Van der Waals forces between the graphite layers. In addition, it is thought that due to the high strength of graphite, the polyamide 6 + graphite composites have increased load carrying capacity and therefore improved wear life. Graphite is thought to reduce the contact temperature during sliding due to its high thermal conductivity and reduce friction by reducing the contact area [29].

For the wax-added polyamide 6 composites, there are no deep wear marks on either the disc or the pin surfaces, and the surfaces are smoother, Figure 8. The reason for this is thought to be due to the bonding effect of the wax on the wear surface, minimizing ruptures and having a solid lubricant effect. Thus, the mass loss in the material decreased and the wear resistance of the polyamide 6 + graphite + wax composites increased. This is consistent with previous literature and is due to the formation of a more homogeneous transfer film layer on the opposite disc surface during wear [22, 30]. Moreover, the decrease in wear rate can be expressed as follows. The addition of graphite to Polyamide 6 polymer decreases degree of crystallinity of the material but increases the stiffness and modulus of elasticity of the composite. In other words, graphite additive increases the stiffness of the composite. In addition, the addition of graphite/ wax solid lubricant additives to the main matrix decreases the stiffness and hardness slightly. In short,

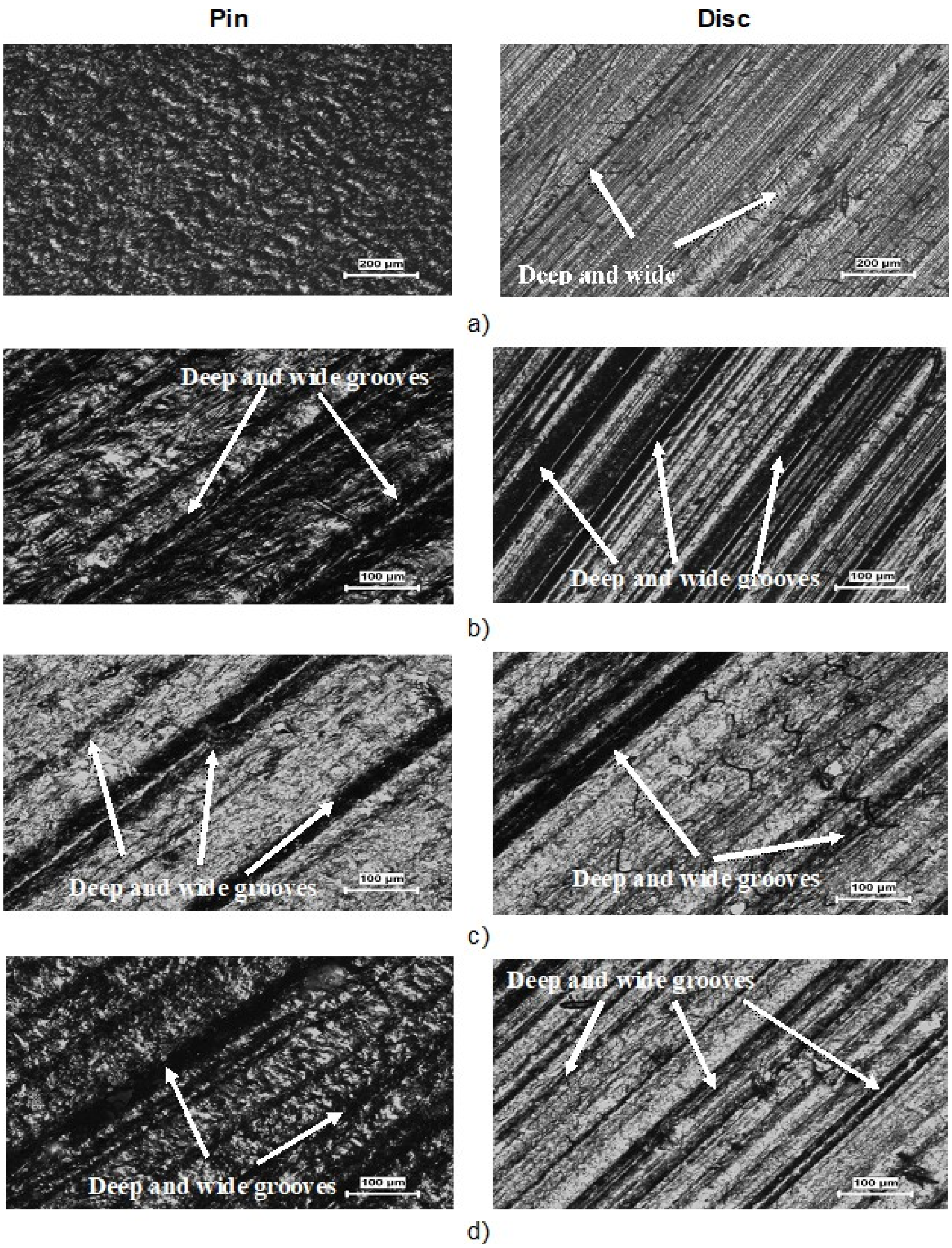

FIGURE 7 Optical microscope images of the worn pin and disc surfaces of a) the pure polyamide 6, b) the polyamide 6–5 % graphite, c) the polyamide 6–10 % graphite, d) the polyamide 6–15 wt.% graphite (sliding speed: 0.5 $m s^{-1}$, load:100 N).

the decrease in the wear rate can also be explained by the increase in the film layer formed on the disc, hardness and tensile modulus. The adhesive wear mechanism is effective for the polyamide-graphite-wax hybrid composite.

4 | CONCLUSIONS

The following results were obtained in this study on the tribological properties of polyamide 6 and polyamide 6/graphite/wax composites.

1. The friction coefficient of the pure polyamide 6 was higher along the sliding distance and the peaks in the friction coefficient-sliding distance graph were longer when compared to the polyamide 6/grahite composites. In addition, it was determined that the friction coefficient value varied along the sliding distance. The friction coefficient values for the polyamide 6/graphite composites were lower and the peaks in the friction coefficient-sliding distance graph were shorter.
2. The friction coefficient values for the polyamide 6/graphite/wax composites were lower than the values

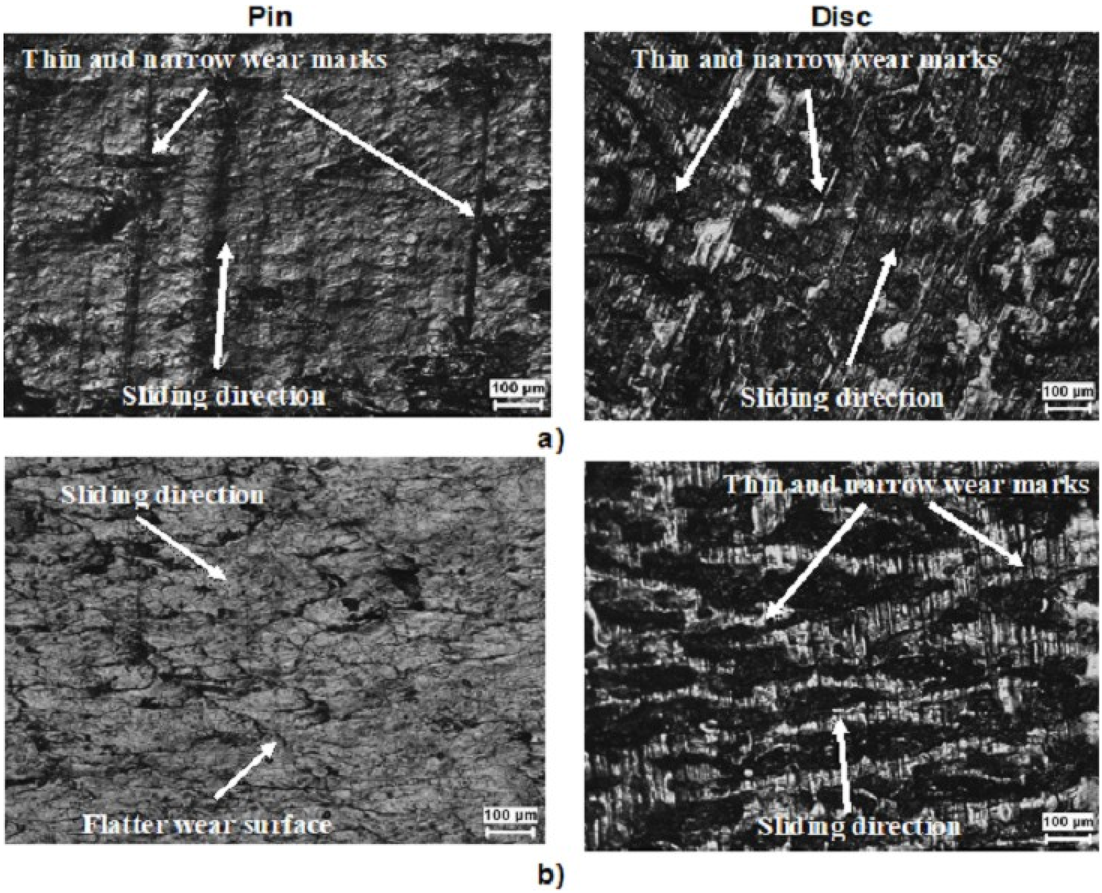

FIGURE 8 Optical microscope images of the worn pin and disc surfaces of a) the polyamide 6–15 % graphite-2 % wax, b) the polyamide 6–15 % graphite-6 % wax (sliding speed: 0.5 m s^{-1}, load:100 N).

for the pure polyamide 6 and the polyamide 6/graphite composites. Although there was a slight tendency for the friction coefficient values of the polyamide 6-graphite-wax to decrease along the sliding distance, no significant change was detected.

3. The friction coefficient value decreased with the increase in the applied load for all composites. The highest coefficient of friction was measured as 0.98 under 10 N load, and the lowest coefficient of friction was measured as 0.07 under 250 N load for the polyamide 6–15 % graphite-6 % wax composite.
4. While the wear rate of the pure polyamide 6 and the polyamide 6-graphite composites increased with increasing load, the wear rate of the polyamide 6/graphite/wax composites decreased.
5. When the optical microscope images of the pure polyamide 6 used in the wear tests were examined, many deep wear marks were observed on the pin surface. The polyamide 6-graphite composites also had wear marks, but the surfaces are brighter.
6. There were no obvious wear marks on either the disc or pin surfaces of the polyamide 6-graphite-wax composites, and it was determined that ruptures, separation and cracks did not occur.
7. The polyamide polyamide 6/graphite/wax composite, produced by the addition of wax to the graphite-added polyamide 6 composite, had a synergetic effect in terms of tribology, and the specific wear rate was reduced by approximately 20 % and 30 %, respectively, when compared to the wear rate values for the pure polyamide 6 and polyamide 6–15 graphite composites, while the friction coefficient values decreased by approximately 75 % and 84 %, respectively.
8. The maximum working load for the pure polyamide 6 and the polyamide 6/graphite composites was 30 N when they worked on themselves at a sliding speed of 0.5 m/s, while the maximum working load of the polyamide 6-graphite-wax polymer composites, produced by adding 2 % and 6 % wax to the polyamide 6-graphite composite, was 250 N when they worked on themselves, which means the addition of wax leads to a load-carrying capacity which was 7.33 times greater.

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